

synthetic

mixed STEM responses

$$Q1 \quad u = x^2, \quad du = 2x \, dx$$

$$I = \frac{1}{2} \int_0^1 e^u \, du = \frac{1}{2}(e - 1)$$

$$Q2 \quad \mu = 0.20 \quad N = mg \cos 30^\circ$$

$$a = g(\sin 30^\circ - \mu \cos 30^\circ) = 3.20$$

$$v = \sqrt{2as} = 4.38 \, \text{m/s}$$

$$Q3 \quad \text{If } A^T A x = 0, \text{ then } A x = 0$$

because A^T cancels A .

Q4

$$m = \max(xs)$$

$$z = [\exp(x-m) \text{ for } x \text{ in } xs]$$

$$\text{return } [v / \text{sum}(z) \text{ for } v \text{ in } z]$$